Surv 617 HW2

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Study objectives

1. On average, how does systolic blood pressure change over time as people get older?

2. Are there differences in systolic blood pressure trajectories by sex?

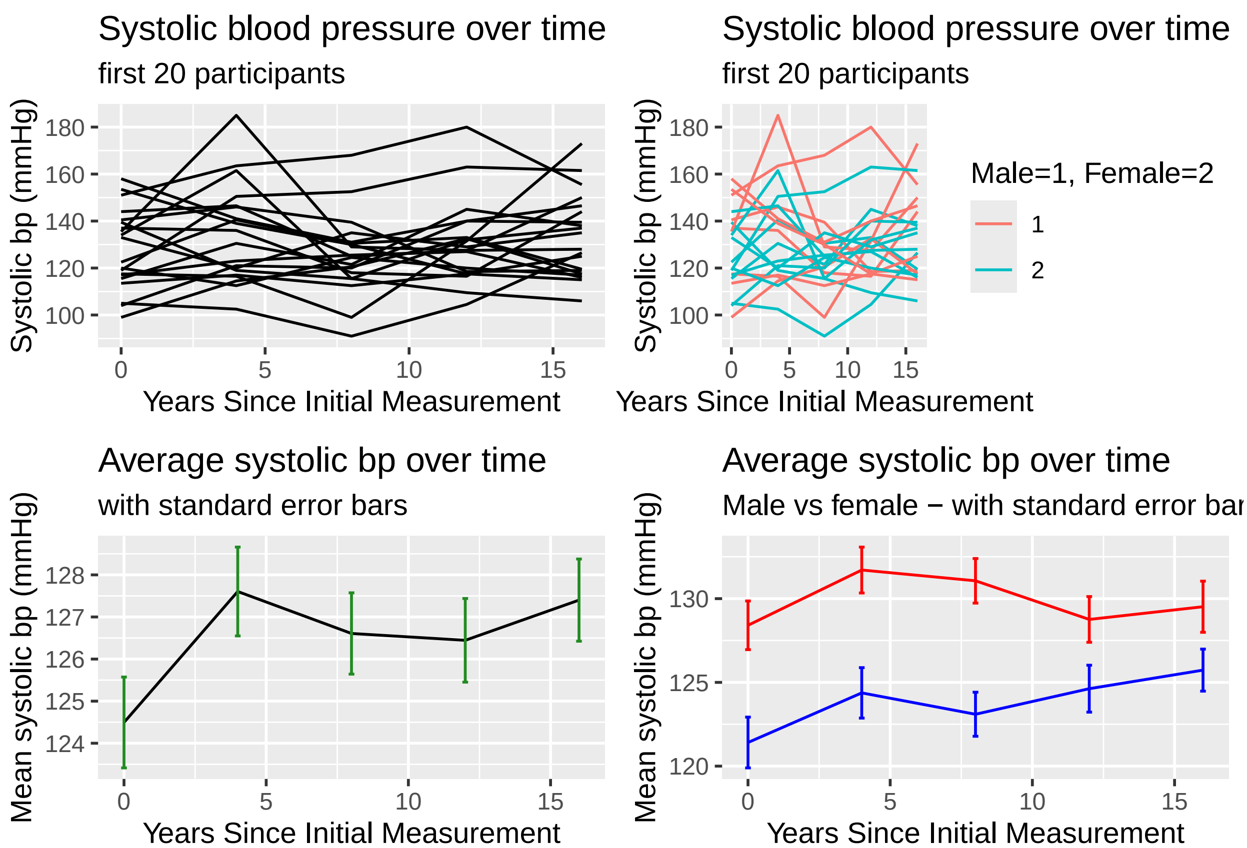
Questions

1.

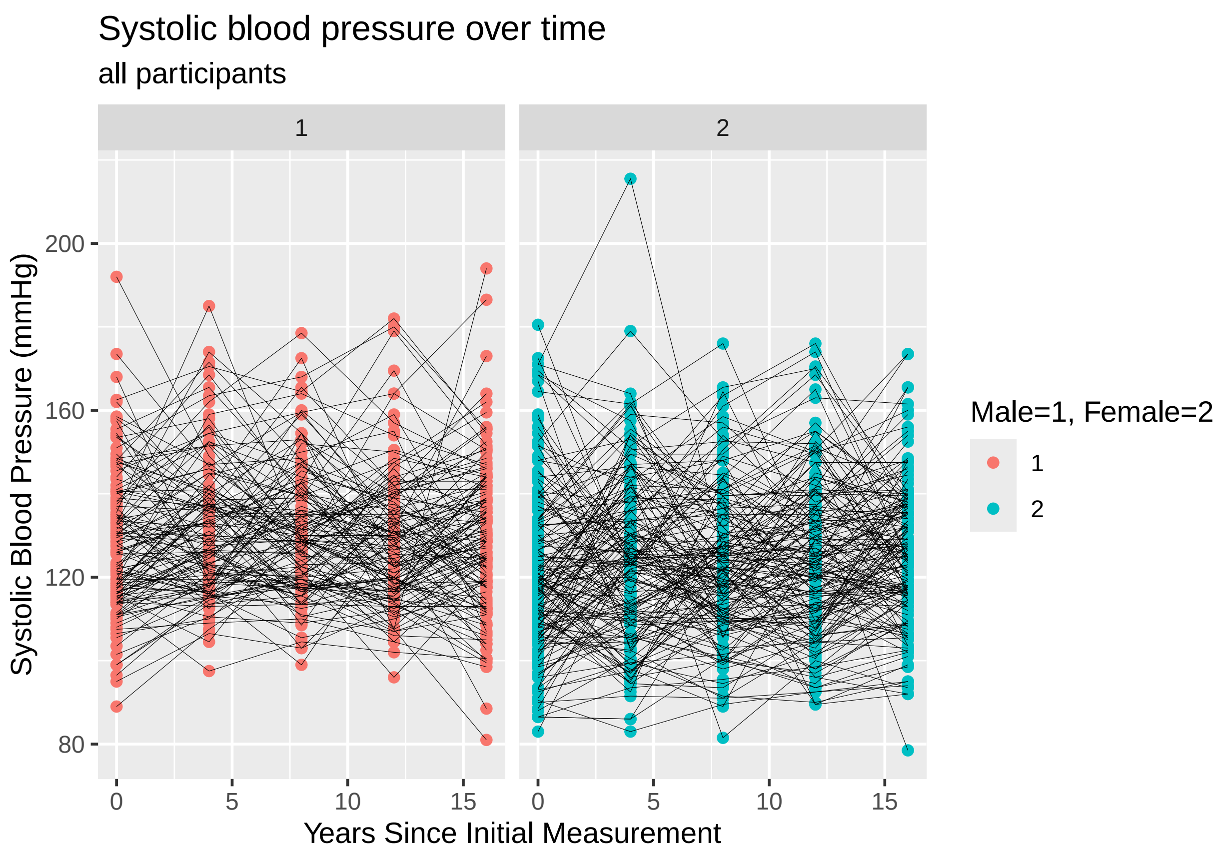
Explore the trajectory of systolic blood pressure over time (both individually and on average) by creating data visualizations. Do you observe evidence of non-linear trends over time? Do you observe evidence that trajectories (i.e., the slopes) differ by sex?

```
library(tidyverse)
library(ggplot2)
library(cowplot)
library(lme4)
library(nlme)
library(geepack)
load("C:/Users/jacma/OneDrive/school work/UMD/SURV 617/Data/hrs_wbbp_long.rda")
hrs_wbbp_long$RAGENDER <- as.factor(hrs_wbbp_long$RAGENDER)

n_part <- 20
men_avg <- subset(hrs_wbbp_long,hrs_wbbp_long$RAGENDER==1) %>%
  group_by(year) %>%
  summarize(avg_bp=mean(bp_sys),
            bp_se=sd(bp_sys)/sqrt(137)) # 137 men
women_avg <- subset(hrs_wbbp_long,hrs_wbbp_long$RAGENDER==2) %>%
  group_by(year) %>%
  summarize(avg_bp=mean(bp_sys),
            bp_se=sd(bp_sys)/sqrt(174)) # 174 women

plot_grid(ggplot(hrs_wbbp_long[1:(n_part*5),]) + ## individual, no gender, subset
            geom_line(mapping=aes(x=year,y=bp_sys,group=HHIDPN)) +
            labs(title="Systolic blood pressure over time",
                 subtitle=paste("first",n_part,"participants",sep=" "),
                 x="Years Since Initial Measurement",y="Systolic bp (mmHg)"),
```

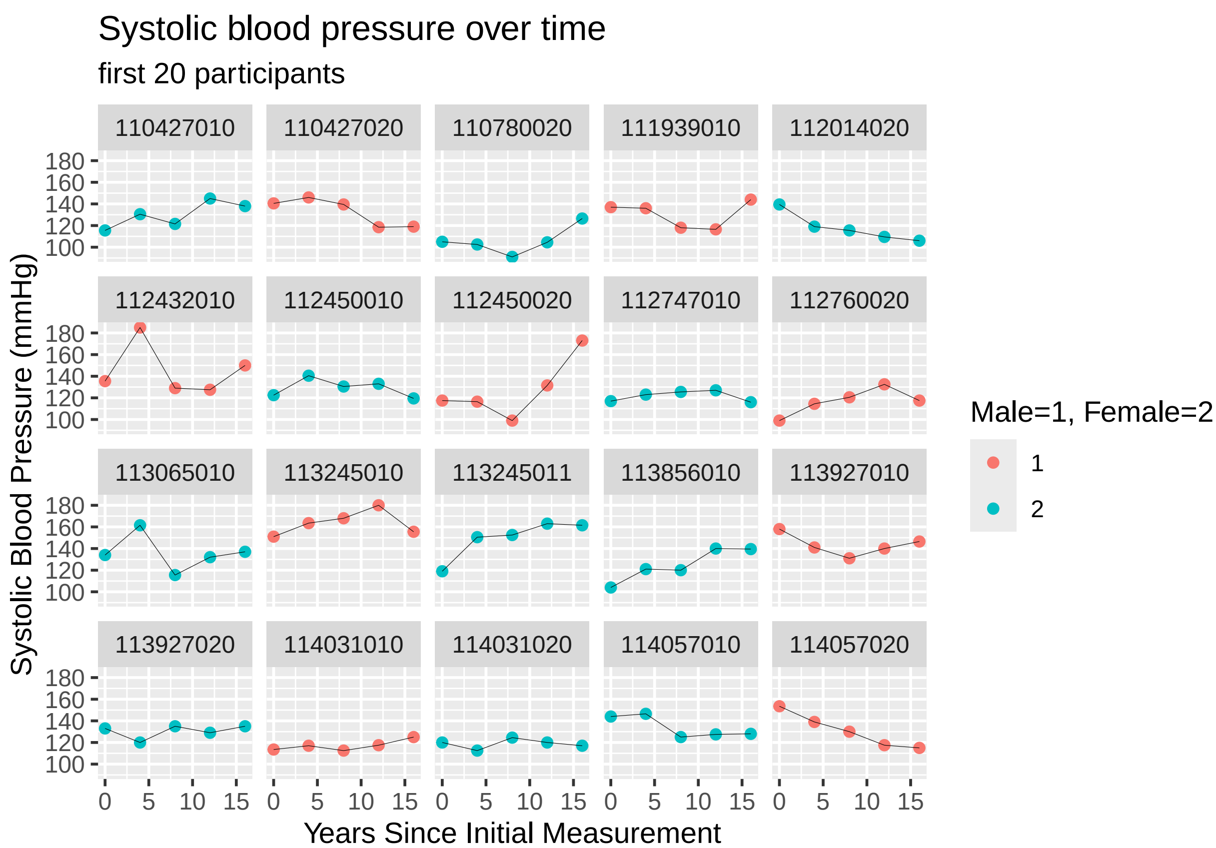
```r
ggplot(hrs_wbbp_long[1:(n_part*5),]) + ## individual, with gender, subset
  geom_line(mapping=aes(x=year,y=bp_sys,group=HHIDPN,color=RAGENDER)) +
  labs(title="Systolic blood pressure over time",
       subtitle=paste("first",n_part,"participants",sep=" "),
       x="Years Since Initial Measurement",y="Systolic bp (mmHg)",
       color="Male=1, Female=2"),

hrs_wbbp_long %>% ## on average, no gender
  group_by(year) %>%
  summarize(avg_bp=mean(bp_sys),
            se_bp=sd(bp_sys)/sqrt(311)) %>% # average computed over each participant
  ggplot() +
  geom_line(mapping=aes(x=year,y=avg_bp)) +
  geom_errorbar(aes(x=year,y=avg_bp,
                      ymin = avg_bp - se_bp, ymax = avg_bp + se_bp),
                width=0.2,color='forestgreen') +
  labs(title="Average systolic bp over time",
       subtitle="with standard error bars",
       x="Years Since Initial Measurement",y="Mean systolic bp (mmHg)"),

ggplot() +
  geom_line(data = men_avg,
            mapping=aes(x=year,y=avg_bp),color='red') +
  geom_errorbar(data=men_avg,
                mapping=aes(x=year,y=avg_bp,
                            ymin = avg_bp - bp_se,
                            ymax = avg_bp + bp_se),
                color='red',width=0.2) +
  geom_line(data = women_avg,
            mapping=aes(x=year,y=avg_bp),color='blue') +
  geom_errorbar(data=women_avg,
                mapping=aes(x=year,y=avg_bp,
                            ymin = avg_bp - bp_se,
                            ymax = avg_bp + bp_se),
                color='blue',width=0.2) +
  labs(title="Average systolic bp over time",
       subtitle="Male vs female - with standard error bars",
       x="Years Since Initial Measurement",y="Mean systolic bp (mmHg)",
       color="Male=1, Female=2"),
nrow=2,ncol=2)
```




```r
# view all data, split by gender
ggplot(data=hrs_wbbp_long, mapping=aes(group=HHIDPN)) +
  geom_point(mapping=aes(x=year,y=bp_sys,color=RAGENDER)) +
  geom_line(mapping=aes(x=year,y=bp_sys),size=0.1) +
  labs(title="Systolic blood pressure over time",
       subtitle="all participants",
       x="Years Since Initial Measurement",y="Systolic Blood Pressure (mmHg)",
       color="Male=1, Female=2") +
  facet_wrap(~RAGENDER)
```




```
# isolate individuals
ggplot(data=hrs_wbbp_long[1:(n_part*5),], mapping=aes(group=HHIDPN)) +
  geom_point(mapping=aes(x=year,y=bp_sys,color=RAGENDER)) +
  geom_line(mapping=aes(x=year,y=bp_sys),size=0.1) +
  labs(title="Systolic blood pressure over time",
       subtitle="first 20 participants",
       x="Years Since Initial Measurement",y="Systolic Blood Pressure (mmHg)",
       color="Male=1, Female=2") +
  facet_wrap(~HHIDPN)
```




```r
rm(n_part,men_avg,women_avg)
```

Looking at individuals
There is certainly variation in the trajectories of blood pressure over time, which can be seen in any of the non-aggregated plots. The trajectories appear to vary in nature (i.e. linear vs quadratic or polynomial), slopes and intercepts, and the size of jumps (some participants, like 112431010, have very large gaps in successive measurements, others are very flat, like 114031010). There does not appear to be an across-the-board difference in *trajectories* between genders. Both men's and women's trajectories exhibit a great deal of variability, in the same ways listed previously. This is best illustrated by the final plot, which shows "constant" (114031020) and "varied" (113255011) women, along with "constant" (114031010) and "varied" (114057020) men. Further, some trajectories decrease (114057020) while others increase (112450020), along with the flat trajectories as noted. There may be some utility in using non-linear relationships with time to capture the variety in trajectories seen in the last plot. It is clear that not all of them are linear - take for instance participants 112432010, 112450020, and 113065010 who had considerable increases *and* considerable decreases. It should also be noted that the reduction to examining only the first "n_part" participants means a great deal of variability is likely missed by these visualizations. One might expect the inclusion of more (or other) participants to further demonstrate the observations discussed here, though. The plot of *all* participants does not seem to suggest that the variability in trajectories would stabilize if more participants were looked at individually - it is very noisy with a great deal of overlapping and criss-crossing lines in the center.

Looking at aggregated data
While the trajectories (slopes) of blood pressure over time do not seem to be deterministic given gender differentiation, there is a clear difference in the intercepts (starting blood pressure) of participants depending on their gender. This is indicated by the second "Average…" plot, which shows men to have higher blood

pressures on average than women, regardless of time point. This may be quite significant because there is no overlap with the respective standard error bars (included to retain some information after aggregation). In other words, gender is likely a key predictor of initial blood pressure (objective 2). While this may be, the change in blood pressure over time according to gender is less distinct. The line segments connecting each time point have similar slopes (in direction and magnitude) for both men and women. That is, the lines appear almost parallel everywhere aside from the year 8 to 12 segment. Overall, one would expect blood pressure to increase over time (judging by the first "Average…" plot which shows a slight upward trend, a pattern that can be seen in the gender-controlled plots as well since both men and women have a net increase in blood pressure at the end of the study), regardless of sex.

2.

Create a variable for age of each participant at baseline (make sure that your intercept is interpretable) and fit a marginal linear model with bp_sys as the dependent variable and fixed effects for age at baseline, RAGENDER, and year. Then fit the same model, but add a quadratic polynomial for year in one model, poly(year, 2), and a cubic polynomial for year in another, poly(year, 3). Use an exchangeable correlation structure for all models. Which model best fits the data?

```r
hrs_wbbp_long <- hrs_wbbp_long %>%
  group_by(HHIDPN) %>%
  mutate(base_age = age_y[which.min(year)]) %>%
  ungroup()
# standardize base_age to make more interpretable (# scale by 311 participants, not 1555 rows)
hrs_wbbp_long$base_age_scaled <- (hrs_wbbp_long %>%
                                    group_by(HHIDPN) %>%
                                    slice(1))$base_age %>%
  scale() %>% rep(each=5)

mlm_1 <- gls(bp_sys~ base_age_scaled + RAGENDER + year,
             correlation = corCompSymm(form= ~ 1 | HHIDPN),
             weights = varIdent(form = ~ 1 | year), # weights?
             data=hrs_wbbp_long)
mlm_2 <- gls(bp_sys~ base_age_scaled + RAGENDER + poly(year,2),
             correlation = corCompSymm(form= ~1 | HHIDPN),
             weights = varIdent(form = ~ 1 | year), # weights?
             data=hrs_wbbp_long)
mlm_3 <- gls(bp_sys~ base_age_scaled + RAGENDER + poly(year,3),
             correlation = corCompSymm(form= ~ 1 | HHIDPN),
             weights = varIdent(form = ~ 1 | year), # weights?
             data=hrs_wbbp_long)

cbind(AIC=c(AIC(mlm_1),AIC(mlm_2),AIC(mlm_3)),
      BIC=c(BIC(mlm_1),BIC(mlm_2),BIC(mlm_3)),
      ResidSE=c(mlm_1$sigma,mlm_2$sigma,mlm_3$sigma),
      LogLike=c(mlm_1$logLik,mlm_2$logLik,mlm_3$logLik))
```

```
##           AIC      BIC  ResidSE   LogLike
## [1,] 12969.55 13023.02 18.49829 -6474.777
## [2,] 12952.50 13011.30 18.47316 -6465.249
## [3,] 12942.95 13007.09 18.46533 -6459.474
```

```r
rm(mlm_1,mlm_2) # remove models not in use anymore
summary(mlm_3)$tTable
```

```
##                      Value  Std.Error    t-value      p-value
## (Intercept)     128.752417  1.0917332 117.933956 0.000000e+00
## base_age_scaled   3.370439  0.7383809   4.564635 5.398793e-06
## RAGENDER2        -4.043454  1.4849321  -2.722989 6.542016e-03
## poly(year, 3)1   26.068831 13.8760016   1.878699 6.047326e-02
## poly(year, 3)2  -15.827584 13.5221563  -1.170493 2.419828e-01
## poly(year, 3)3   28.714157 13.5129916   2.124930 3.375042e-02
```

Model 3, with a quadratic and cubic term for the effect of time on blood pressure, had the lowest AIC, BIC, residual standard error, and largest log-likelihood. For these reasons, one can claim it fits the data the best of the three options. This result makes some sense, because it was established that some of the trajectories were not monotonic (in that they decreased then increased, some even alternating more than once). The inclusion of higher-order polynomial terms allows the model to capture a bit of that variability in the trajectories ("shapes") of blood pressure over time.

3.

For the model you selected in Q2, interpret all non-polynomial regression coefficients (regardless of whether they are significant) from the model fit in plain English. Also, given the estimated polynomial coefficients, what do these suggest about the nature of the overall trajectory in blood pressure over time?

- *Intercept*: The intercept is the average value of blood pressure when all other variables are 0 (or reference category). The average blood pressure at time 0, for a male of average (according to the sample) age is 128.75 mmHg. This parameter was extremely statistically significant.
- *base_age_scaled*: The parameter estimate was 3.37, meaning an increase of 1 in the scaled age at baseline (i.e. 1 standard deviation higher, about 4.8 years) of the participant increases systolic blood pressure by 3.37 mmHg, when all other variables are held constant (i.e. at a particular time point, male/female). This parameter was also very significant, with the second-smallest p-value.
- *RAGENDER*: The estimate was -4.04. Since RAGENDER=1 (males) were used as the reference, this parameter being negative suggests that the participant being female decreases their initial blood pressure by this amount, all else held constant (i.e. at any particular time point, age). Like the parameters above, the effect of gender was significant with a very small p-value.
- *Polynomial Coefficients*: All of the polynomial coefficients had large estimates (26.06 for the linear term, -15.83 for the quadratic, 28.71 for the cubic), which seems to suggest that the data is extremely noisy/varied, and the model is potentially overfitting more extreme values, leading to larger parameter estimates. Together, the coefficients build a monotonically-increasing function (on its practical domain, i.e. the slope is always positive when year > 0, it is always positive in general), which suggests that the overall trajectory of blood pressure over time is *increasing*. The cubic term is the only one which is significant at $\alpha = 0.05$, which could suggest that the trajectories follow some sort of cubic trajectory (although the lower-order terms should still be included in models)

4.

Refit the model you chose in Q2, this time using an AR1 working correlation structure in one model, and unstructured (general) in another. Use appropriate methods to compare these models against each other and the model with an exchangeable correlation structure and identify which appears to fit the data best. Examine the estimated variance-covariance matrices for each of the three models and briefly comment on what you observe.

```r
# AIC, BIC to compare
mlm_3_ar <- gls(bp_sys~ base_age_scaled + RAGENDER + poly(year,3),
                correlation = corAR1(form= ~ 1 | HHIDPN),
                weights = varIdent(form = ~ 1 | year), # weights?
                data=hrs_wbbp_long)

mlm_3_unst <- gls(bp_sys~ base_age_scaled + RAGENDER + poly(year,3),
                  correlation = corSymm(form= ~ 1 | HHIDPN),
                  weights = varIdent(form = ~ 1 | year),# weights?
                  data=hrs_wbbp_long)

cbind(AIC=c(AIC(mlm_3),AIC(mlm_3_ar),AIC(mlm_3_unst)),
      BIC=c(BIC(mlm_3),BIC(mlm_3_ar),BIC(mlm_3_unst)),
      ResidSE=c(mlm_3$sigma,mlm_3_ar$sigma,mlm_3_unst$sigma),
      LogLike=c(mlm_3$logLik,mlm_3_ar$logLik,mlm_3_unst$logLik))
```

```
##           AIC      BIC  ResidSE   LogLike
## [1,] 12942.95 13007.09 18.46533 -6459.474
## [2,] 12941.53 13005.67 18.33582 -6458.763
## [3,] 12909.92 13022.18 18.25496 -6433.962
```

It is not completely clear which model is the best, because the unstructured correlation model had the lowest AIC and standard error, but the autoregressive model had the lowest BIC. Either of these may be selected, although more intuition is given below.

```r
getVarCov(mlm_3)
```

```
## Marginal variance covariance matrix
##        [,1]   [,2]   [,3]   [,4]   [,5]
## [1,] 340.97 130.66 119.31 123.07 128.91
## [2,] 130.66 316.34 114.92 118.54 124.16
## [3,] 119.31 114.92 263.77 108.24 113.38
## [4,] 123.07 118.54 108.24 280.65 116.95
## [5,] 128.91 124.16 113.38 116.95 307.90
##   Standard Deviations: 18.465 17.786 16.241 16.752 17.547
```

```r
# corMatrix(mlm_3$modelStruct$corStruct)[[1]] # exchangeable - all entries same
getVarCov(mlm_3_ar)
```

```
## Marginal variance covariance matrix
##         [,1]    [,2]    [,3]    [,4]    [,5]
## [1,] 336.200 161.070  69.838  35.239  17.454
## [2,] 161.070 328.170 142.290  71.795  35.561
## [3,]  69.838 142.290 262.360 132.380  65.570
## [4,]  35.239  71.795 132.380 284.050 140.700
## [5,]  17.454  35.561  65.570 140.700 296.360
##   Standard Deviations: 18.336 18.115 16.197 16.854 17.215
```

```r
# corMatrix(mlm_3_ar$modelStruct$corStruct)[[1]] # further-away time points less correlation
getVarCov(mlm_3_unst)
```

```
## Marginal variance covariance matrix
##         [,1]    [,2]    [,3]   [,4]    [,5]
## [1,] 333.240 154.600 106.340 102.67  98.854
## [2,] 154.600 321.060 131.850 132.44  93.588
## [3,] 106.340 131.850 267.230 155.42  80.214
## [4,] 102.670 132.440 155.420 291.97 136.230
## [5,]  98.854  93.588  80.214 136.23 294.110
##   Standard Deviations: 18.255 17.918 16.347 17.087 17.15
```

```r
# corMatrix(mlm_3_unst$modelStruct$corStruct)[[1]] # every entry unique

rm(mlm_3,mlm_3_ar,mlm_3_unst)
```

Of course, the exchangeable structure (first matrix) would have the same estimated correlation for each off-diagonal entry (by definition) when standardized. The auto-regressive structure has decreasing covariances for later time points, as expected, while there does not seem to be a clear pattern in the unstructured covariance matrix (naturally, because each entry is determined uniquely). The standard deviations of the diagonals are all *extremely* similar, and so are the model fits, but the unstructured specification has the best diagnostics (for AIC and residual standard error, at least), and thus may be the model of choice. Of note is that the computational time of the last model is by far the longest due to its unstructured correlation matrix. This may be relevant for much larger data sets.
Generally, since all of the model fit criteria are similar, it may even be reasonable to defer to the analysts' belief on the nature of the data when it comes to selecting the optimal correlation structure. My intuition is that AR(1) may not make the most sense, because if a participant has a very high blood pressure at the start of the study, I wouldn't expect their blood pressure at the next wave to be less correlated to that initial value than their blood pressure at the final wave. That is, a participant with high blood pressure "just is" a participant with high blood pressure, and from the EDA at the beginning it seems that net-negative trajectories are pretty rare. I am also wary of the unstructured correlation, both because it does not seem intuitive that the relationship between time points is different depending on the points selected, *and* because it makes interpretation quite tricky. An exchangeable structure feels the most appropriate because it suggests the correlation between blood pressure for any two time points is the same. This structure also feels more "person-specific" rather than "time-specific" like the others, which again relates to what the precise research goal is.

5.

Individuals can be diagnosed with hypertension (high blood pressure) when their systolic blood pressure is greater than or equal to 140. Create a variable called htn that equals 1 when bp_sys >=140 and 0 otherwise. Then fit a multilevel logistic regression model with htn as your dependent variable and year as your predictor (not as a polynomial). Include a random intercept and a random year slope for each individual. Interpret the fixed effects on both the log-odds and odds ratio scales.

```r
hrs_wbbp_long$htn <- ifelse(hrs_wbbp_long$bp_sys>=140,1,0)
hyper_1 <- glmer(htn ~ year + (year|HHIDPN),
                 family = binomial("logit"), data = hrs_wbbp_long)
summary(hyper_1)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: htn ~ year + (year | HHIDPN)
##    Data: hrs_wbbp_long
```

```
##
##      AIC      BIC   logLik deviance df.resid
##   1499.4   1526.1   -744.7   1489.4     1550
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.2536 -0.4392 -0.2581 -0.2472  2.1932
##
## Random effects:
##  Groups Name        Variance Std.Dev. Corr
##  HHIDPN (Intercept) 2.551615 1.59738
##         year        0.004363 0.06605  -0.18
## Number of obs: 1555, groups:  HHIDPN, 311
##
## Fixed effects:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.87173    0.24183  -7.740 9.95e-15 ***
## year        -0.00641    0.02342  -0.274    0.784
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## year -0.750
```

- *Intercept*: The overall log-odds of a person having hypertension at baseline (year=0) is -1.8717, meaning the probability of blood pressure over 140 at the start of the study is less than 0.5. The corresponding odds ratio is 0.1538, which means that the odds of having hypertension at baseline are about 85% lower than the odds of *not* having hypertension. This is corroborated by the fact that only about 20% of participants had blood pressure above 140 at the start of the study. This parameter is capturing most of the variability for this model, since it is extremely significant.
- *Year*: The log-odds of a person having hypertension decreases by 0.00641 each year, which corresponds to an odds ratio of 0.9936. The former means that for every year that passes (i.e. increase by 1), the log odds of hypertension are expected to decrease by 0.00641, and latter means that we expect to see a decrease of about 0.6% in the odds of hypertension each year. This suggests that the odds of hypertension barely changes as time goes by. The parameter estimate is very statistically insignificant which supports this interpretation and implies that initial blood pressure has a much greater effect on the odds of hypertension than time itself.

6.

Refit your model from Q5, adding RAGENDER and baseline age as fixed effects.

```r
hyper_2 <- glmer(htn ~ year + RAGENDER + base_age_scaled + (year|HHIDPN),
                 family = binomial("logit"), data = hrs_wbbp_long)
summary(hyper_2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: htn ~ year + RAGENDER + base_age_scaled + (year | HHIDPN)
##    Data: hrs_wbbp_long
```

```
##
##      AIC      BIC   logLik deviance df.resid
##   1489.8   1527.3   -737.9   1475.8     1548
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.3639 -0.4318 -0.2635 -0.1956  2.8316
##
## Random effects:
##  Groups Name        Variance Std.Dev. Corr
##  HHIDPN (Intercept) 1.912293 1.38286
##         year        0.005019 0.07084  0.11
## Number of obs: 1555, groups:  HHIDPN, 311
##
## Fixed effects:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -1.59461    0.24975  -6.385 1.72e-10 ***
## year            -0.01959    0.02266  -0.865  0.38729
## RAGENDER2       -0.32847    0.24963  -1.316  0.18823
## base_age_scaled  0.39773    0.14255   2.790  0.00527 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) year   RAGEND
## year        -0.577
## RAGENDER2   -0.473 -0.079
## base_g_scld -0.146 -0.175  0.286
```

```r
hyper_2 %>% resid() %>% var()
```

```
## [1] 0.6368388
```

7.

Interpret the model parameters estimated in Q6, including any estimated variance components. Be sure to interpret all fixed effects (regardless of significance) on both the log-odds and odds ratio scales.

- *Intercept*: Log-odds of -1.5946 mean that the probability of hypertension at baseline (start of study) for an average-aged (about 59) male is less than 0.5. The odds ratio of 0.203 means that the odds of this individual having hypertension are about 80% lower than not. In other words, most individuals do not have hypertension at baseline.
- *year*: When year increases by 1, the log odds of hypertension decreases by 0.0196. That is, the odds of hypertension are *lower* as year increases (the odds decrease over time). The odds-ratio conversion gives a value of 0.9806, meaning the odds of hypertension are 2% lower from one year to the next. This effect is for any fixed value for gender or baseline age. While the result does not seem intuitive, the parameter is not significant at all which means it may be "ignorable".
- *RAGENDER*: The estimate of -0.3284 for log-odds means that women are less likely to have hypertension than men. Specifically, by converting to odds ratio, their odds are about 28% lower than men (1-exp(-0.3284)), all else held constant. This parameter is not significant though, suggesting hypertension may be due in larger part to other factors (like the following). This is the effect at any fixed value for year or age.

- *base_age_scaled*: An estimate of 0.3977 here means that for every 1 unit increase in scaled age at baseline (i.e. around 4.8 years), the log-odds of hypertension increases by 0.3977. In other words, the odds of having hypertension are greater for older participants. In terms of odds ratio, a participant one standard deviation (4.8 years) older than the average participant has odds of hypertension about 48.8% higher than average-aged participants, when all other variables are held constant (at a given time point, male/female). Two standard deviations higher would have an increase in odds of about 97.6% (almost twice as likely), and so on.
- *Variance in random intercepts*: 1.912 means there is some variability in the odds of hypertension at baseline across different participants. The initial blood pressure is intuitively very predictive of whether or not the participant will have (or already may have) hypertension, and there is considerable variation in these starting values, so it makes sense for most of the variation to be in this random effect.
- *Variance in random slopes*: The small value of 0.005 suggests that there is little variation in the trajectory of hypertension odds over time, matching the interpretation of the insignificant parameter estimate for "year". The odds of hypertension could be said to be relatively constant over time and consistent across participants (by reconciling this result and the one for "year"). The word "consistent" here means that the trajectory of hypertension odds is similar across all participants, and "constant" means that the trajectories themselves do not have large slopes. The key takeaway is that the odds of hypertension are determined moreso by the participant's baseline age.
- *Correlation between random slopes and random intercepts*: 0.11 means that the random slopes and random intercepts are not very strongly correlated. So, a participant with a very high starting blood pressure is not necessarily expected to have a steeper *or* flatter trajectory of hypertension odds over time. The same is true for participants with very a low starting blood pressure.
- *Variance in residuals*: The leftover variability not explained by the model parameters is 0.6368, which seems to be fairly small.

8.

Fit a marginal logistic regression model with htn as the dependent variable and year, RAGENDER, and baseline age as fixed effects. Use an exchangeable working correlation structure. Interpret the regression coefficients on either the log-odds or odds ratio scale. Compare your estimates and interpretations to the multilevel model fit in Q6.

```r
# gee - fitting marginal models to non-normal dependent variables
hyper_gee <- geeglm(htn ~ year + RAGENDER + base_age_scaled, id=HHIDPN, waves=year,
                    family=binomial("logit"), data=hrs_wbbp_long, corstr="exchangeable")
summary(hyper_gee)
```

```
##
## Call:
## geeglm(formula = htn ~ year + RAGENDER + base_age_scaled, family = binomial("logit"),
##     data = hrs_wbbp_long, id = HHIDPN, waves = year, corstr = "exchangeable")
##
##  Coefficients:
##                  Estimate   Std.err   Wald Pr(>|W|)
## (Intercept)     -1.1700553  0.1568238 55.666 8.59e-14 ***
## year            -0.0004826  0.0101570  0.002   0.9621
## RAGENDER2       -0.2645401  0.1838891  2.070   0.1503
## base_age_scaled  0.2649036  0.1110803  5.687   0.0171 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation structure = exchangeable
## Estimated Scale Parameters:
```

```
##
##               Estimate Std.err
## (Intercept)    1.003  0.1039
##   Link = identity
##
## Estimated Correlation Parameters:
##       Estimate Std.err
## alpha   0.2495 0.04316
## Number of clusters:   311  Maximum cluster size: 5
```

- *Intercept*: Log-odds of -1.17 mean that participants with hypertension at baseline are less common than those without it. The corresponding odds-ratio of 0.31 means that the odds of hypertension at baseline (year=0) for an average-aged (about 59) male are 70% lower than not having hypertension.
- *year*: Log-odds of -0.000483 mean that as year increases by 1, the average overall log-odds of having hypertension are expected to decrease by a VERY small amount. The odds-ratio is almost exactly 1, meaning the odds of hypertension are basically the exact same from one year to the next (on average, with the marginal model). This suggests that the overall slope of hypertension odds is relatively constant with respect to time.
- *RAGENDER*: An estimate of -0.2645 means that women have lower log-odds of hypertension than men, on average. In terms of odds ratio, womens' odds are about 24% lower than men (1-exp(-0.2645)). This matches all previous findings, where women were found to have significantly lower blood pressures overall, and were consistently lower than men on the visualizations at the beginning of this document. This parameter is not significant, though, probably because the following has a greater effect.
- *base_age_scaled*: The parameter estimate of 0.2649 here means that an increase in scaled age at baseline by one unit (about 4.8 actual years of age) increases the log-odds of hypertension by 0.2649. An odds-ratio of 1.3, one would say that being one standard deviation above the mean age increases the odds of hypertension by 30%, two standard deviations above the mean increases them by 60%, etc. Generally, the older the participant is, the more likely they are to have hypertension

```r
rbind(GLM=hyper_2 %>% fixef(),
      GEE=hyper_gee$coefficients)
```

```
##     (Intercept)       year RAGENDER2 base_age_scaled
## GLM      -1.595 -0.0195895   -0.3285          0.3977
## GEE      -1.170 -0.0004826   -0.2645          0.2649
```

```r
rm(hyper_1,hyper_2,hyper_gee,hrs_wbbp_long)
```

The estimates are the same in terms of direction, but GEE attenuates each coefficient (i.e. brings them closer to 0). GEE captures whether the overall marginal differences in the probability of hypertension (across different values of the explanatory variables) changes over time. That is, inference is made at the population level rather than the individual level. The interpretations of the parameters are the same on their own, GEE just finds a lesser effect for each variable.

9.

In ~1/2 page, describe the models fit in Q6 and Q8. How are they similar and how are they different? In what situations would you choose one over the other?

Both models use year (time), RAGENDER (sex), and base_age_scaled (standardized age at the start of the study) as independent variables for the binary dependent variable hrt (1 if blood pressure is greater than or equal to 140, 0 otherwise). Question 6 fits a logistic multilevel model (MLM - a hierarchical model), while Question 8 fits a Generalized Estimating Equation (GEE - a marginal model). These are similar in that they both include fixed effects (namely for the independent variables listed previously), but the primary distinction is that the second model *does not include random effects.* This is reflective of the primary situation in which one would be chosen over the other. If the researcher intends to study the variability in the trajectories hypertension odds over time across participants, the hierarchical model would be selected because it includes random effects (i.e. allowing the effect of time on hypertension odds to vary between individuals). The random effects allow the variability in intercepts and slopes (i.e. the trajectories) to be examined. In this instance, it was found that there is very little variability in the trajectory of hypertension odds among participants (in other words, their chances of having hypertension over time are mostly similar to each other). On the other hand, if the researcher is interested in determining if the overall marginal differences in log odds change over time with different levels of the explanatory variables, they would opt for GEE. This research objective does not hold interest for the variability between individuals. Inference for GEE is at the *population level*, contrasting to multilevel model's inference at the *individual level*.

Reframing to the study objectives at the beginning of the document: #1 is best answered with a GEE because it is interested in the *average* change in blood pressure over time, while #2 could be explored with either multilevel or GEE depending on whether the researcher is interested in if the effect of sex on blood pressure trajectory varies among individuals (MLM), or if they simply want the overall effect of sex on blood pressure trajectory (GEE).

As far as interpretations are concerned, the MLM and GEE are very similar (with fixed effects, of course, since they are incomparable in regard to random effects since GEE does not even include them). Both estimate increases in year and the participant being female to decrease the log-odds of hypertension, while estimating an increase of baseline age to increase the log-odds of hypertension. Both have negative intercepts, meaning the odds of having hypertension for an average-aged male at the start of the study are much lower than not having hypertension. GEE's coefficients are smaller, suggesting a lesser effect. More analysis could be done on the significance of the parameters, but the primary motivation in choosing which effect to focus on comes from the research objectives as discussed.

Also worth noting (in a more general sense) is that GEE may be preferred for other reasons such as it being more robust - the coefficients are consistent regardless of the selected correlation structure). It may also be preferred if there is a very large number of clusters, because multilevel models can be extremely computationally expensive in such a scenario. However, multilevel models are *the* choice if there is any interest in variability among clusters in the data.

Note: Residual diagnostics were not done in this document since they were not prompted, and take up quite a bit of space. qq-plots, histograms of random effects, and heteroskedasticity could all be easily checked for each model (where applicable).